Semantics of Causal Probabilistic Programming: Interventions, Equations and Cycles

Qualifying Dissertation for the Transfer of Status

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Abstract

Causal inference asks what would happen if a variable were forced to a value, and the standard answer, Pearl's structural causal models, is a syntactic transformation of a model. Causal probabilistic programming aims to make such interventions primitives of an expressive probabilistic language, and existing languages do so by evaluation strategies without an account of what they compute. In my first year, with coauthors, I gave that account: a hierarchy of probabilistic languages in which interventions and counterfactual twinning are program transformations, causal structure is defined by a contextual equivalence of programs, and a graded denotational semantics is fully abstract for it and extends to higher order. The proposed thesis builds outward from this base in the three places where causal models stop—the equational theory and its relation to factorisation, cyclic models as fixpoints, and abstraction between models—and makes the result practical with a vectorised implementation. A second line of work, an algebraic theory of quantum communication in which the order of events is denotational data, is reported as work to date and as the thesis's outlook. Part I reviews the literature the thesis builds on; Part II states the research plan and is self-contained.

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Part I
Literature Review

1 Introduction

Three notions of causality. When a computer scientist says that one part of a system *causes* another, they may mean at least three distinct things. In Pearl's theory of causal inference [105], X has a causal effect on Y if *intervening* on X—forcing it to a value, regardless of how it would otherwise have been produced—changes the distribution of Y. In quantum foundations and in the theory of higher-order quantum processes [80, 100], the causal structure of a process is given by its *signalling* constraints, which inputs can influence which outputs, together with the further question of whether it is compatible with a definite causal order between its parts. In probabilistic modelling, a causal structure is a directed acyclic graph along which a joint distribution *factorises*, and the content of the structure is the family of conditional independences it entails [105, 53].

Three further usages should be named so as to be set aside. In categorical quantum mechanics, a process is called *causal* when it preserves the discarding map [40]. This is the causality axiom of Chiribella, D'Ariano and Perinotti [33], glossed as "no signalling from the future", and Coecke [39] shows that it implies the no-signalling principle for processes composed along a causal structure; it is the base axiom on which signalling structure is built, not a notion of causality between parts. Fritz [52] calls a Markov category *causal* when the output of a morphism cannot depend on a choice made downstream of it; this is again a property of the ambient category. In artificial intelligence and in verification, *actual causality* [63] asks which particular event caused which particular outcome in a single run; that question is not treated here. Nor are experimental tests of indefinite causal order, causal discovery, or identifiability, each of which is mentioned below only where a work package touches it.

That the same word carries these meanings is not an accident, but neither is it a theorem. Each notion is *extensional*: it characterises causal structure by what an observer may do to the system—intervene, send signals, condition. What such observers probe is *intensional*: the dataflow of a program, the temporal order of its communication events, the graph of a model. Programs

carry this intensional structure explicitly in their syntax. The viewpoint of this thesis is therefore that causality is a property of *program semantics*, and that the right way to relate the three notions is to relate, for each one, the intensional structure in the syntax to the extensional characterisation in the semantics by a correspondence theorem.

Where the two routes stand. Two lines of work precede this proposal, and the review is organised around them. On the interventional side, my coauthors and I built a hierarchy of first-order probabilistic languages in which Pearl's interventions and counterfactual *twinning* are program transformations, defined program equivalences contextually, and gave a graded denotational semantics that is fully abstract for concrete models (§3). This route has the complete picture for one notion of causality: the intensional dataflow of a program is characterised extensionally by equivalence under all interventional contexts. On the communication side, my MSc dissertation gave an algebraic theory $\mathsf{IO} \otimes \mathsf{Q}$ of classically controlled quantum communication, an operational model in which programs exchange qubits with a quantum stream, and a denotational model given by a free I/O monad on $\mathbf{CP}^{\infty}$, with the denotational model adequate and fully abstract for the operational one; this year's work, with my supervisor, is its conference write-up and a draft proof of completeness (§4). Here the temporal order of communication events is first-class semantic data: the denotation is indexed by I/O traces. This route does *not* yet have an extensional causal reading of that order; giving it one, as signalling structure in the sense of higher-order quantum processes, is part of the research plan rather than a result.

A single example shows why these are two faces of one question. In the interventional language, a program reads an *intervention label* ℓ through $\mathsf{applyIntv}\ \ell\ t$, which yields the label's value when it is set and t otherwise; if a second variable z does not depend on the result, every intervention at ℓ leaves the distribution of z unchanged, and the contextual equivalence makes this an equation between programs. Causal independence is *tested by contexts*. In the theory $\mathsf{IO}\otimes\mathsf{Q}$, the commutation axiom $\mathsf{apply}_U(\boldsymbol{a},\ \boldsymbol{a}.\ \mathsf{in}(b.\ x(\boldsymbol{a},b))) = \mathsf{in}(b.\ \mathsf{apply}_U(\boldsymbol{a},\ \boldsymbol{a}.\ x(\boldsymbol{a},b)))$ says that a local operation and a communication event on different qubits may be reordered, and the quantum-stream semantics validates it; by contrast in and out do not commute with one another, and that is exactly why the trace index is a total order. Causal independence is *an equation validated by an observer*. Both statements say that no information flows from one place to another, so that reordering or intervening is unobservable; in programming-language terms both are *noninterference* properties [60]. Whether this coincidence is an instance of a general correspondence is the question of Part II.

Scope of the review. Sections 2–4.3 treat the literature on which the thesis builds: the semantics of effects, probabilistic programs and quantum programs; causal models, their equivalences and their diagrammatic presentations; causal probabilistic programming; and quantum process calculi. Sections 4.4 and 4.5 position two neighbouring fields that the research plan reaches into, the semantics of concurrency and higher-order quantum processes, at the level of their main definitions and results; their detailed survey is a milestone of that plan (Part II, §12). Section 5 draws the three notions together and identifies the gap the thesis addresses.

2 Semantic foundations shared by both routes

2.1 Monads, algebraic theories, and their combinations

Moggi's observation that notions of computation are monads [95, 96] and Plotkin and Power's refinement that most of them are presented by algebraic theories of operations and equations [109, 110] are the common foundation of both routes. An algebraic theory determines a monad, and conversely a finitary monad is a Lawvere theory [69]; the equational presentation is what makes program equivalence a matter of derivability. Hyland, Plotkin and Power [68] showed that the standard ways of combining theories have computational meaning: the *sum* adjoins the operations of two theories with no interaction, so that the order of effects from the two theories is observable, while the *tensor* additionally imposes that operations of one theory commute with those of the other,

so that their relative order is unobservable. In the premonoidal semantics of effects [112, 113, 55], the morphisms that commute with everything are the *central* ones; centrality is the established algebraic form of "cannot be observed to happen before or after". Handlers [14] give the programming counterpart of models; comodels and runners [111, 6] describe the environments in which effectful programs execute.

Ordinary algebraic theories cannot express operations that create and consume *resources*, such as qubits, which by no-cloning must be treated linearly. Staton's *parameterised algebraic theories* (PATs) [134, 135] extend the framework with parameters: operations consume parameters, classically choose a branch, and bind fresh parameters in each branch. Staton used them to give a fully complete equational theory of qubit quantum computation [135]; Stark's free model of π-calculus name creation [133] is the earlier instance of the same commutation discipline, where communication primitives commute with independent local operations. Kammar, Liell-Cock, Lindley, Matache and Staton [76] have given a complete equational axiomatisation of dynamic threads (fork, wait, stop, atomic actions) as a parameterised algebraic theory, whose closed terms correspond exactly to pomsets; I return to this in §4.4, since it is the nearest neighbour of my own completeness result.

Where we stand. My MSc dissertation [141] shows that sums and tensors exist for the enriched Lawvere theories presented by PATs, by explicit syntactic construction, which is what licenses the definition of $\mathsf{IO} \otimes \mathsf{Q}$ as a tensor in §4.

What this gives us. The tensor, and centrality more generally, are already primitive statements about causal independence. In $\mathsf{IO} \otimes \mathsf{Q}$ the tensor makes communication commute with local operations on other qubits, while in and out do not commute with each other. I treat this as an observation to be made precise, not as a result: Part II asks (WP 3) whether centrality in an algebraic theory is in general the algebraic shadow of no-signalling.

2.2 Semantics of probabilistic programs

Probabilistic programming gives a compositional syntax for statistical models and performs Bayesian inference on them compositionally [91]. Its denotational semantics is monadic: the Giry monad on measurable spaces [81, 136], and, for higher-order programs, the probability monad on quasi-Borel spaces [64]. Implementations such as LazyPPL [42], MonadBayes [123, 124] and GenJAX [15] realise this monadic structure directly. Full abstraction is the recurring theorem shape for such languages, the standard reference point being probabilistic coherence spaces for probabilistic PCF [49].

The synthetic approach of *Markov categories* [52, 54] axiomatises the categorical structure needed to talk about probability without measure theory: a symmetric monoidal category with a commutative comonoid on every object, where discarding is natural (the unit is terminal) but copying need not be. In Markov categories with conditionals, disintegration and Bayesian inversion are expressed synthetically [36, 52], and Fritz and Klingler [53] prove the soundness of d-separation, and the equivalence of the local and global Markov properties, synthetically; completeness of d-separation in the sense of Geiger, Verma and Pearl remains the classical result. This is the *factorisation* face of causality: a Bayesian network is a string diagram in a Markov category, and the conditional independences it entails are read off the diagram.

The second piece of structure both routes rely on is *grading*. Graded monads [78, 98, 56, 46] and, more generally, locally graded categories [87, 121] track an index—an effect annotation, a trace, a resource—through composition. Milius, Pattinson and Schröder [92] use graded monads for trace semantics; Liell-Cock and Staton's **ImP** construction [88] uses grading to make imprecise probability compositional. The interventional semantics of §3.5 is built on an analogous construction in which the grade records which interventions are available.

What this gives us and what it lacks. Markov categories give factorisation but no primitive notion of intervention; the diagrammatic causal-model literature (§3.3) adds it by surgery on diagrams. Neither offers a notion of program *context*, which is what contextual equivalence needs.

2.3 Semantics of quantum programs

Selinger's QPL [125] established the denotational template for quantum programming languages: types denote finite-dimensional spaces of density matrices (or, more generally, C^*-algebras [35]), and programs denote completely positive maps. Categorical quantum mechanics [3, 65, 40] gave this a graphical calculus through dagger compact closed categories and the CPM construction [126], and the ZX-family of calculi axiomatise fragments of it; Staton's PAT Q [135] and Carette et al.'s extension of Π [27] give complete equational theories of qubit programs and of unitary circuits respectively. Languages such as QWIRE, Quipper and Silq [104, 61, 19] work in the circuit model with classical control. For languages with infinitary features one needs the completion of **CP** under infinite biproducts, $\mathbf{CP}^\infty$ [102, 150], which is the base category of the communication semantics. Quantum monads [122, 8] and quantum game semantics [37] are two further lines of work on the structure of quantum computation as an effect; the latter uses event structures and is the closest existing semantics to the one proposed in WP 2a.

What this gives us. The axiom that singles out the deterministic (discard-preserving) processes in categorical quantum mechanics [40, 33] is the base condition from which the signalling structure of §4.5 is built; the **Caus** construction requires compact closure and further "precausality" axioms on top of it, and, by design, its base category also contains the non-deterministic, trace-decreasing processes.

3 Causality as intervention

3.1 Structural causal models and Pearl's hierarchy

Pearl's structural causal models (SCMs) [105, 106] are systems of equations $X_i = f_i(\mathrm{Pa}_i, U_i)$ in which each endogenous variable is a deterministic function of its parents and of an exogenous noise variable; a prior on the noise makes an SCM a probabilistic model, and the equations make it more than a distribution. The surplus is organised by the *causal hierarchy* [11]: rung 1 asks observational questions; rung 2 asks *interventional* questions, answered by replacing the equation for X with a constant (the do operator); rung 3 asks *counterfactual* questions about a particular run, answered by abduction, action and prediction, implemented graphically by *twinning*, which duplicates the model into worlds that share the noise [10]. Both operations are *syntactic transformations of the model*. The Neyman–Rubin potential-outcomes framework [120] is the other dominant formalism; its correspondence with SCMs is well understood [72]. Richardson and Robins' single-world intervention graphs [117] present an intervention by splitting a node into its natural and its fixed value, which is the graphical counterpart of the intervention labels of our languages; they are deliberately single-world, and cross-world quantities such as natural direct effects need further assumptions there. Interventions need not be perfect: the literature on stochastic and soft interventions [43, 44, 48, 86] replaces an equation by another mechanism rather than a constant.

Throughout, I work with fully specified models and Bayesian inference: mechanisms and priors are given, and queries are answered by inference. *Identifiability* (when an interventional quantity is computable from observational data, the subject of the do-calculus [128, 127]) and *causal discovery* [132, 108] are outside the scope of this thesis.

What this gives us. The causal structure of an SCM is its dataflow: which variables each equation reads. Interventions probe it. The semantic question is when two models have the same causal structure, and the answer must be extensional.

3.2 Equivalences of causal models and logical accounts

Bongers, Forré, Peters and Mooij [23] give the principal account of equivalence for SCMs at all three rungs: two models are observationally, interventionally or counterfactually equivalent when they agree on the corresponding class of queries, with the counterfactual case defined through the two-world twinned model. Their treatment also covers cyclic SCMs, through solution functions. At rung 2, several further notions exist: Pearl's causal Bayesian networks are families of interventional

distributions compatible with a graph; *causal spaces* [103, 25] axiomatise this measure-theoretically; and *causal abstraction* [119, 16, 118, 90, 50] studies coarsening maps between models that commute with interventions. At the level of graphs rather than models, Markov equivalence [138] is what discovery algorithms can recover. On the logical side, Ibeling and Icard [71] characterise the three rungs by three probabilistic logics of strictly increasing expressivity whose semantics is defined on SCMs through Pearl's operations, and Icard has proposed probabilistic Turing machines as the computational counterpart of SCMs [73, 70].

What this gives us and what it lacks. These are all extensional characterisations of interventional structure, and they are the yardstick against which our contextual equivalences are measured (§3.5). None of them is formulated as a contextual equivalence of programs, and none treats higher-order models.

3.3 Diagrammatic and categorical causal models

Fong's *causal theories* [51] present a Bayesian network as a symmetric monoidal category freely generated by its graph, with models as functors into stochastic maps. Jacobs, Kissinger and Zanasi [74] present interventions as *string-diagram surgery* in this setting, and introduce *comb disintegration*: the use of combs, processes with a hole, to express the classical causal-inference computations, which is the first appearance of the comb notion of §4.5 in classical causal inference. Lorenz and Tull [89] develop causal models in string diagrams systematically, and a later line presents causal abstraction as natural transformations between such models [90, 50]. Bayesian inference in this style is handled by *normalisation boxes*, which renormalise locally inside a diagram [75], and Agrawal et al. [5] obtain equational principles for causal Bayesian networks from them. Zhang et al. [149] characterise d-separation as a noninterference-style separation property, which makes explicit a connection implicit in all of this work and in §1: invariance under interventions is a noninterference property [60].

What this gives us and what it lacks. This is the closest existing bridge between the factorisation face and the intervention face of causality, and it is compositional in the functorial sense. Its equational theories are, however, stated per graph, and it has no notion of program context or of higher-order models. It is where the unifying work package (WP 3) starts, and comb disintegration is the prior work that WP 3a must be positioned against.

3.4 Causal probabilistic programming languages

Causal probabilistic programming aims to make intervention and twinning primitives of an expressive probabilistic language. Libraries such as DoWhy [116, 22] and the `do` operators of Pyro and PyMC [20, 4] manipulate models as graphs or computation graphs and stop at rung 2 (DoWhy's GCM module reaches rung 3 on fitted mechanisms). Witty et al. [146, 147] treat an intervention as a source transformation that replaces subterms, which is powerful but non-compositional; Brulé's Whittemore [24] is an early embedded causal DSL.

Four systems reach rung 3 natively. OmegaC [137] obtains interventions by a lazy evaluation strategy that retroactively modifies closures, reminiscent of dynamic binding, and counterfactuals by making every program a deterministic function of explicit noise. ChiRho [30] implements both as effect handlers over Pyro, with a multi-world value type whose relation to Pearl's twinning is not immediate. MultiVerse [107] uses trace inspection and re-execution. Kiesel et al. [79] answer counterfactual queries on ProbLog programs by a transformation proved correct against Pearl's semantics.

What this gives us and what it lacks. These systems show that the primitives are implementable at higher order, by non-standard evaluation techniques. What they lack is a denotational and equational foundation: a semantics in which intervention is a mathematical operation rather than an evaluation strategy, and an account of when two causal programs are equivalent. That is what the first route supplies.

3.5 Where we stand on the interventional route

Our POPL submission [FILL: status: under revision for POPL 2027?; coauthors] (an early version appeared at the LAFI workshop [140]; artifact [139]) reconstructs Pearl's hierarchy as a hierarchy of three first-order probabilistic languages. L_1 is a standard first-order probabilistic language with conditioning by a final score. L_2 exposes hooks in the dataflow as typed free variables, the *intervention labels*, read through $\mathsf{applyIntv}$; an intervention $\mathsf{do}(\ell \mapsto v)\{t\}$ is substitution of v for ℓ. L_3 annotates each binding as exogenous or endogenous, and *twinning* twin_n is a vectorisation transformation from L_3 to L_2 that duplicates the endogenous bindings into n worlds sharing the exogenous ones. Program equivalence at rung k is *contextual* with respect to rung-1 denotational equality, in Milner's sense: two programs are $\equiv_k$ when they are denotationally equal in every rung-k context, and varying the admissible interventions and world counts gives a spectrum of equivalences. For representable acyclic SCMs compiled into L_3, the observational, interventional and counterfactual equivalences of Bongers et al. coincide with the equivalences $\equiv_3^{\text{obs}}$, $\equiv_3^{\text{intv}}$ and $\equiv_3^{\text{2-world}}$ of the compiled programs; the full $\equiv_3$ quantifies over all n-world twinnings, and whether it is strictly finer than $\equiv_3^{\text{2-world}}$ is open. The denotational semantics interprets interventions by *grading*, in a construction reminiscent of **ImP** [88], and twinning by world-polymorphic vectorisation. It is adequate for the contextual equivalences unconditionally and fully abstract when the underlying model is *concrete*, meaning that values correspond to global elements and global elements separate morphisms; **Set**, standard Borel spaces and quasi-Borel spaces are concrete, and the construction is parametric in the model. The denotation then motivates a higher-order language L_{HO} with cross-world interventions, into which L_3 compiles in a semantics-preserving way, and which has a definitional Haskell implementation. [FILL: one sentence on which parts are yours: e.g. the contextual-equivalence framework and the full-abstraction proofs]

The limitations relevant to Part II are these: the languages are acyclic; the relation between the contextual equational theory and the normalisation-box equational theories of §3.3 is not worked out; the first-order semantics is over normalised probability, while L_{HO} widens to the unnormalised measure monad to internalise conditioning, so that programs with observations denote sub-normalised maps; and the models are measure-theoretic rather than finite-dimensional, which matters for the signalling face (§5).

4 Causality as order and signalling in communicating quantum programs

4.1 Quantum communication and distributed quantum computing

Quantum communication underlies teleportation [18], key distribution [17] and, increasingly, proposals for distributed quantum computing that connect small quantum processors into larger ones [26]. Programming models for distributed quantum computing are at present circuit-level: they describe which gates are applied where and which qubits are sent when. On the semantic side, Ceragioli, Gadducci, Lomurno and Tedeschi observe that the existing quantum process calculi do not agree on what the observational properties of quantum values should be [29]. There is, in other words, no agreed account of what a distributed quantum program *means*, nor of which of its behaviours are physically realisable.

4.2 Quantum process calculi

Quantum communication has mostly been studied through quantum process calculi—CQP [57], QPAlg [83], qCCS [148] and lqCCS [29]—which extend classical calculi [93, 66, 94] with qubit message passing. They are presented operationally, by labelled transition systems over configurations, with program equivalence given by bisimilarity. As the authors of lqCCS show, the proposed bisimilarities differ in subtle ways because of the interaction of concurrency, synchronisation and local nondeterminism with quantum effects; their later coalgebraic account of effect-labelled transition

systems [28] is an attempt to organise these subtleties. A separate difficulty is well known from the classical theory: parallel composition is not an algebraic operation and cannot be adjoined to an algebraic theory as an operation [133, 59], although workarounds exist, such as the cooperative threads of Abadi and Plotkin [1].

What this gives us and what it lacks. Process calculi are the only existing semantic treatment of communicating quantum programs, and they are expressive. They are operational only, their equivalences are not canonical, and they entangle communication with concurrency, which makes it hard to isolate what is specific to *communication*. My work studies communication in isolation, as an algebraic effect, and is in that sense the local sublanguage of these calculi with a denotational semantics.

4.3 Where we stand on the communication route

My MSc dissertation [141], and its conference version with my supervisor [FILL: FoSSaCS submission status], introduce $\mathsf{IO} \otimes \mathsf{Q}$, an algebraic theory of classically controlled qubit communication, as the *tensor* of a standard I/O theory [68] with Staton's complete theory Q [135]: programs may send and receive qubits depending on measurement outcomes, and communication commutes with local quantum operations. The tensor is argued to be needed, since one equation that holds in the tensor but not in the sum is required to validate a quantum phenomenon involving entanglement across the program–environment boundary (Example B in Part II). The theory is given two models. The operational model lets a program exchange qubits with an environment modelled as a *quantum stream*; a program equivalence $\approx_q$ compatible with quantum theory is defined from it by inspecting final quantum states, and terms modulo $\approx_q$ form a model of $\mathsf{IO} \otimes \mathsf{Q}$, which shows that the tensor does not collapse in the manner of Eckmann–Hilton. Although operational, $\approx_q$ is therefore not a contextual equivalence; whether a contextual equivalence observing only traces, continuations and a final measurement coincides with it is a question I return to in Part II. The denotational model is the free I/O monad $T(X) = \bigoplus_\sigma \mathrm{qbit}^{|\sigma|} \otimes X$ on $\mathbf{CP}^\infty$, indexed by I/O traces σ. The main theorem is that the denotational semantics is adequate and fully abstract for the operational one: $[\![c]\!] = [\![c']\!]$ iff $c \approx_q c'$. What the dissertation left as a conjecture is *completeness*: that $[\![t]\!] = [\![t']\!]$ implies $t = t'$ is derivable in $\mathsf{IO} \otimes \mathsf{Q}$. I now have a draft proof of this (manuscript in preparation, not yet closed), by induction on the I/O depth of terms, using a reorganisation step that normalises every term into an I/O-free head applied to a single input and a single output branch, and a factorisation lemma on completely positive maps that lets semantically equal terms be rewritten to a common prefix. If it closes, the three semantics—algebraic, operational, denotational—coincide.

The observation that sets up Part II is this. The trace σ is a *temporal* order of communication events, exposed in the denotation: the σ-component of a closed term is a completely positive map with $|\sigma|$ open qubit wires, one per event, to be used in the order σ, and when the term branches on measurements this component is sub-normalised, its trace being the probability of the branches that produce σ. Whether the family of these components is a *causal* object in the signalling sense of §4.5, a comb or a tester whose slots are ordered by σ, is a question the dissertation raised and did not answer. It is stated as Conjecture 1 in Part II.

4.4 Order in concurrency theory

Concurrency theory has long modelled causal order explicitly. Lamport's happens-before relation [84] is the partial order an observer of a distributed system can reconstruct. Mazurkiewicz traces [45] are words modulo the commutation of independent letters, the cheapest partial-order model and the one that arises when independent channels are added to a sequential program. Pratt's pomsets [114]—partially ordered multisets of events, up to isomorphism, with series and parallel composition—generalise words to partially ordered words, and Gischer [58] axiomatised the equational theory of series-parallel pomsets. Winskel's event structures [144, 97, 145] add a conflict relation to the causal order and give a domain-theoretic account of nondeterministic concurrent computation. Concurrent Kleene algebra [67] adds a commutative parallel operator to Kleene al-

gebra, related to sequential composition by an exchange law; Kappé, Brunet, Silva and Zanasi [77] prove that series-parallel pomset languages closed under Gischer's subsumption order are its free model, by reduction to an earlier completeness theorem for bi-Kleene algebra [85]. Within the algebraic-effects tradition, Dvir, Kammar and Lahav [47] give a monadic semantics of shared-state concurrency from an equational theory, and Kammar et al. [76] axiomatise dynamic threads as a parameterised algebraic theory whose closed terms are exactly pomsets, with a completeness theorem. Clairambault, de Visme and Winskel [37] give an event-structure game semantics for quantum programs.

Position. The trace-indexed monad of §4.3 is the degenerate single-channel case of this picture: a single channel forces a total order, so traces are words. Two extensions are distinguished in Part II. Adding independent channels to a *sequential* program makes events on different channels commute, so that traces become Mazurkiewicz traces and the monad a free monad over a partially commutative monoid; this keeps the proof architecture of the single-channel case. Adding *threads* makes traces pomsets, and the theory of Kammar et al. is a parameterised algebraic theory with exactly that free model, so "tensor $\mathsf{IO} \otimes \mathsf{Q}$ with a thread theory" is a well-posed, and much harder, second step. Completeness results in this literature also go through free models; whether their techniques transfer to the quantum setting, where a factorisation lemma on CP maps did the work, is part of the first reading milestone of the plan. The logics of session types are the other point of contact: BV and its relatives have been used as logics of multiparty protocols, and Simmons conjectures that causal consistency of a wiring of black boxes corresponds to deadlock freedom of the protocol [129]; a trace-indexed semantics of communicating programs is where such a correspondence could be made precise.

4.5 Higher-order quantum processes and signalling

Higher-order quantum computation studies transformations of quantum processes rather than of states. Chiribella, D'Ariano and Perinotti's *quantum combs* [31, 32] are processes with holes in a fixed causal order; their probabilistic counterparts, *testers*, and Gutoski and Watrous's *strategies* [62] describe multi-round interactions with sub-normalised outcomes, and the general theory of supermaps forms a hierarchy of types [21]. Oreshkov, Costa and Brukner's *process matrices* [100] are the most general bipartite process operators compatible with quantum mechanics holding locally in each party's laboratory; the multipartite theory and the notion of *causal separability*, in which the order of later parties may depend classically on earlier outcomes, are due to Araújo et al. and Oreshkov and Giarmatzi [9, 101]. Some process matrices are causally non-separable. Two kinds of evidence distinguish them: device-independent violation of a *causal inequality*, which Oreshkov, Costa and Brukner's process exhibits, and device-dependent witnesses of non-separability. The *quantum switch* [34], which applies two operations in an order controlled by a qubit in superposition, is causally non-separable [9] but produces only causal correlations; photonic experiments implement it [115], and whether such implementations realise or simulate indefinite order is debated [99, 142].

Which indefinite-order processes can be built as circuits is the subject of Wechs, Dourdent, Abbott and Branciard [143], who define nested classes of quantum circuits with fixed order, with *classical* control of causal order (the next operation may depend on classical outcomes of earlier ones) and with *quantum* control of causal order, and characterise each by conditions on process matrices. The switch lies in the quantum-control class and not in the classical-control one; every classical-control process is causally separable; and quantum-control processes generate only causal correlations. That the quantum-control class exhausts the physically realisable processes is a proposal, not a theorem.

Categorically, Kissinger and Uijlen's **Caus** construction [80] builds, over a *precausal* category—compact closed, with a chosen discarding structure and further axioms, the examples being **CP** on finite-dimensional systems and $\mathbf{Mat}(\mathbb{R}_{\geq 0})$, finite matrices of non-negative reals—a $*$-autonomous category whose types are signalling constraints: first-order processes, non-signalling processes, one-way signalling processes, and so on. The dual of the non-signalling type contains all bipartite process matrices, causally separable or not; separability is a property inside that type, not a type, and

Simmons proves that every process matrix is an affine combination of definite-order processes [129]. Simmons's thesis [129] is the fullest development of this line and the reference for most of what Part II needs on the signalling side. It rebuilds the construction over *additive* precausal categories (compact closed with products, hence biproducts, and a finite causal basis for every object), which gives **Caus** products, coproducts and probabilistic choice as type operators; it adds a non-commutative sequential operator for one-way signalling, under which **Caus** models Guglielmi's BV-logic [130]; it defines n-ary *graph types* encoding compatibility with an arbitrary DAG of signalling relations; it surveys candidate ways of admitting sub-normalised, partial processes into the construction, effectus-style, and where each falls short; and it introduces *causal logic*, pomset logic with directed axiom links, whose proof nets characterise exactly the causally consistent wirings of black boxes [129, 131], with BV and pomset logic separated by a process-matrix example. The finite-causal-basis axiom excludes infinite-dimensional and measure-theoretic bases, which is one of the two obstructions discussed in §5. Finally, *quantum causal models* transfer Pearl's framework to quantum processes: Costa and Shrapnel [41] gave the first process-matrix-based definition, Allen et al. [7] the common-cause line, and Barrett, Lorenz and Oreshkov [12] the general framework, in which a node is an input–output pair, the process operator factorises as a product of commuting operators (a quantum Markov condition), interventions are local instruments at nodes, and d-separation in the graph implies conditional independence of classical outcomes. The cyclic extension [13] gives a cyclic-causal-model account of some processes without definite order, including the switch. Coherent control of channels [2, 82] and, on the language side, the PBS-calculus of Clément and Perdrix [38], which has the switch as a term, are the relevant prior work on coherent control.

Position. Three things follow for this thesis. First, the family of σ-components of an $\mathsf{IO} \otimes \mathsf{Q}$ denotation has the shape of a tester or strategy whose slots are ordered by the traces; that it *is* one is Conjecture 1. Second, signalling types are the extensional face of order in the same way that interventional equivalence is the extensional face of dataflow, and quantum causal models are the one existing place where factorisation, intervention and signalling are all present, which is why they are the first reading milestone. Third, the analogy between parallel composition and indefinite causal order must be stated with care, and the process-matrix picture is the right way to state it: the parties' local programs are combs, and the process matrix is the *network* that wires them together, in other words the scheduler. Parallel composition $P \parallel Q$ leaves the order of events unspecified. A classical scheduler, even one whose later choices depend on earlier outcomes, yields a process in the classical-control class, hence a causally separable one. The switch is a coherent scheduler, well-typed on every non-signalling pair of channels, whose output applies the two operations in a superposition of orders; it is causally non-separable, and it has content only when the two programs act on a shared target or communicate, since on disjoint systems every order gives the same channel. The analogy is therefore between $\parallel$ and the argument type of a scheduler, not between $\parallel$ and the switch itself; this is what makes WP 2d well-posed, and also what makes it hard, since coherent control of order is not an algebraic operation with classical continuations. Finally, Simmons's own outlook [129] proposes reading send and receive operations as physical processes so as to assign causal types to communicating programs, with causal consistency playing the role of deadlock freedom and the session-type logics BV, GV and MAV as the point of contact; that is, from the physics side, the direction of WP 2a and WP 2b.

5 Synthesis: three notions of causality and the gap

Table 1 summarises the review. Each column is one extensional notion of causality; each row asks the same question of it.

Three remarks. First, the example of §1 and the asymmetry it exposes: for intervention the correspondence between intensional and extensional structure is a theorem; for signalling there is a semantics with order as first-class data and a conjecture. Second, non-DAG structure is the next hard case in every column, but it is not the same case: feedback in cyclic SCMs, causal non-separability in quantum processes, and unscheduled parallel composition in programs are three

	Intervention	**Signalling**	**Factorisation**
Intensional datum	dataflow of a program / SCM	temporal order of events: trace σ, comb order, pomset	DAG
Extensional characterisation	contextual equivalence under interventions	signalling types (**Caus**); causal (non-)separability	conditional independence / d-separation
Correspondence theorem	full abstraction for concrete models (§3.5)	full abstraction and draft completeness relate three equalities (§4.3); the signalling reading is Conjecture 1	soundness of d-separation, synthetically [53]; completeness classically
Semantics of programs	first-order and higher-order, acyclic	single channel, classical control	Markov categories; interventions by surgery
Next hard case	cyclic SCMs	multiple channels; causally non-separable processes	cyclic and quantum causal models

Table 1: Three notions of causality and where the two routes stand.

different departures from a fixed order, and a unifying account must distinguish them. Third, and this is the gap: there is no framework in which the three columns are structure on one category. Markov categories cover factorisation and, through surgery, part of intervention; **Caus** covers signalling for quantum and finite classical processes; quantum causal models relate all three, with a compositional presentation in Lorenz and Tull's string diagrams, but are not formulated as a semantics of programs. The two semantics of §3.5 and §4.3 cover the first two columns separately, and both are families of morphisms indexed by an intensional datum, intervention grades or traces, with a reindexing action, which is the common structure a generalisation would start from.

Two obstructions should be displayed rather than hidden. The first is *finite dimension*: the **Caus** construction needs a compact closed base with invertible dimensions, which $\mathbf{Mat}(\mathbb{R}_{\geq 0})$ and finite-dimensional **CP** provide, but measurable spaces, quasi-Borel spaces and the infinite-biproduct completion $\mathbf{CP}^{\infty}$ do not. The bridge of WP 3a is therefore stated on the finite fragment, and the comb reading of WP 2b is stated per trace, over finite-dimensional **CP**, with the trace index kept external. The second is *sub-normalisation*: conditioning in L_{HO} denotes sub-normalised maps, and branching in $\mathsf{IO} \otimes \mathsf{Q}$ makes individual trace components sub-normalised, so neither lands in a first-order **Caus** type as it stands; testers and strategies [32, 62] are the objects that accommodate sub-normalised families, Simmons [129] surveys the ways of admitting partial processes into **Caus** and finds each wanting in some respect, and normalisation boxes [75] are the diagrammatic tool for renormalising after conditioning. I also note that the factorisation column is the thinnest for me: it enters through Markov categories and the diagrammatic causal-model literature, not through my own results, and WP 3a includes a factorisation reading for that reason.

Part II
Thesis Proposal

Notation and background in brief

- *Interventional languages* $L_1 \subseteq L_2 \subseteq L_3$ and L_{HO}: first-order probabilistic languages with intervention labels ℓ read by $\mathsf{applyIntv}$, interventions $\mathsf{do}(\ell \mapsto v)\{t\}$ by substitution, and twinning twin_n as a vectorisation from L_3 to L_2; L_{HO} is the higher-order extension with cross-world interventions. $\equiv_k$: contextual equivalence at rung k with respect to rung-1 denotational equality; $\equiv_3^{\text{obs}}, \equiv_3^{\text{intv}}, \equiv_3^{\text{2-world}}$: the restricted equivalences matching Bongers et al.
- *Graded semantics*: the denotation of an L_2 program is indexed by the interventions available,

in a construction reminiscent of **ImP** [88]; a model is *concrete* when values are global elements and global elements separate morphisms. Full abstraction holds for concrete models.
- *Factorisation*: Markov categories; Bayesian networks as string diagrams; d-separation; normalisation boxes [75] and the equational principles for causal Bayesian networks derived from them [5].
- *Cyclic models*: Bongers et al.'s SCMs with cycles, interpreted through solution functions.
- *Communication theory* $\mathsf{IO} \otimes \mathsf{Q}$ and its trace-indexed semantics (§4.3): work to date, referred to in WP 5 only.

6 Research topic and fundamental challenges

The thesis will develop the *semantics of causal probabilistic programming*: a programming-language account of Pearl's causal models in which causal structure is defined by a contextual equivalence of programs, interventions and counterfactuals are mathematical operations on denotations rather than evaluation strategies, and the account extends to the three places where existing causal models stop: higher order, cycles, and abstraction between models. The finished work of §3.5 establishes the base: the hierarchy of languages, the contextual equivalences and their correspondence with Bongers et al., the graded semantics and its full abstraction, and the higher-order language. The thesis builds outward from it in four directions, each in literature I already command: the equational theory and its relation to the factorisation face (Markov categories), cyclic models as fixpoints, relations between models (abstraction and query logics), and a practical language with a vectorised implementation.

The fundamental challenges are these. (i) *Two equational theories for one object.* The contextual equivalences of programs and the normalisation-box equations of causal Bayesian networks are both equational theories of interventional structure, defined in different ways; relating them is a comparison of a program-level and a graph-level theory. (ii) *Cycles.* Bongers et al.'s cyclic SCMs have semantics only through solution functions under solvability conditions; nothing in the graded semantics handles fixpoints, and feedback interacts with interventions in ways acyclic models never exhibit. (iii) *Abstraction is not an equivalence.* Causal abstraction is a directed relation between models of different granularity; making it a program transformation that commutes with interventions requires the graded semantics to be functorial in the model, not just in the program. (iv) *Scale.* The existing implementation is a definitional interpreter over MonadBayes; realistic causal inference needs vectorised inference, and the type-level guarantees of L_{HO} may not survive the move.

7 Goal

I want a mathematical account of what it means for one part of a program to cause another, in the sense that changing it changes the other, that works for programs with higher-order functions, feedback loops and different levels of description, and that can be run.

8 State of the art and why the problem is hard

The existing notions of equivalence for causal models, Bongers et al.'s hierarchy, causal spaces and causal abstraction, say when two models have the same interventional behaviour, but they are not equivalences of *programs*: there is no notion of context and no higher order. Causal probabilistic programming languages, OmegaC, ChiRho and MultiVerse, implement interventions and counterfactuals at higher order by evaluation strategies, without a denotational or equational account of what they compute. The diagrammatic line, string-diagram surgery and normalisation boxes, has equational principles for causal Bayesian networks but states them per graph, with no notion of program. Cyclic causal models have a semantics only through solution functions. The logical char-

acterisation of the hierarchy by Ibeling and Icard organises queries where the programming-language account organises models, and the two have not been related.

The problem is hard because each extension breaks something the base relies on. Higher order breaks the correspondence with graphs; cycles break the dataflow reading of causal structure; abstraction breaks the fixed-model assumption; and scale breaks the definitional implementation. The base result shows that contextual equivalence survives higher order; the thesis asks whether it survives the other three.

9 Approach

Method. The method is the one that produced the base result. (1) Fix a small language whose *syntax* carries the causal structure: intervention labels in the dataflow. (2) Define an *extensional* equivalence from what an observer may do: quantify over interventional contexts. (3) Build a *denotational* model in which the available interventions are a grade, with reindexing. (4) Prove a *correspondence theorem*, full abstraction, between (2) and (3). Each work package adds one feature to the language, re-derives the equivalence and the grade, and re-proves the correspondence.

What is new. Causal structure is *defined* by a contextual equivalence of programs rather than by a property of a graph, which is what lets it survive higher order, cycles and abstraction; the equational theory of causal models becomes a theory of program equations, comparable with the graph-level theories; and the whole account is executable, so that every semantic definition is also an implementation.

Worked example 1: mediation. In L_2, interventions are exposed as typed free variables ℓ read through applyIntv, which returns the label's value when it is set and the program's own value otherwise:

```
let t  = bernoulli 0.5 in              -- treatment
let t' = applyIntv ℓ t in             -- intervenable copy of t
let m  = bernoulli (f t') in           -- mediator
let y  = bernoulli (g t' m) in         -- outcome
let z  = bernoulli 0.3 in              -- unrelated variable
return (y, z)
```

Under $\mathsf{do}(\ell \mapsto \mathsf{Just\ true})\{-\}$ the variable t' denotes the constant true, and the total effect of t on y is the difference between the distributions of y under the two perfect interventions. The natural direct effect, the effect of t on y with the mediator held at the value it would have taken under $t = \mathsf{false}$, is a cross-world quantity; in L_{HO} it is expressed by twinning the program and intervening on the mediator of one world with a value read from the other. The variable z has the same distribution in every interventional context, and $\equiv_2$ makes this an equation between programs of the same type: the program with `return` z in place of `return` (y, z) is $\equiv_2$ to `let` $t' =$ `applyIntv` ℓ `true in let` $z =$ `bernoulli` 0.3 `in return` z, in which the label is consumed but ignored. Full abstraction says that the graded denotation detects exactly these equations. The same program, drawn as a string diagram in a Markov category, is a causal Bayesian network, and the same independence is a d-separation; WP 1 asks whether the two equational theories agree.

Worked example 2: feedback. Replace the mediator's equation by one that reads the outcome, $m =$ `bernoulli` $(f\ t'\ y)$, so that m and y depend on each other. No acyclic program denotes this; Bongers et al. interpret it through a solution function, a distribution over (m, y) that satisfies both equations when one exists and is unique. In a language with a fixpoint construct `fix` $(m, y).\ (\dots)$ the model is a program again, and the question of WP 2 is which fixpoint construct denotes the solution function, and what intervening on m means when m is defined by a fixpoint: Bongers et al. answer by cutting the equation and re-solving, and the program-level answer must agree with it on the models where both are defined.

Work package 1: the equational theory and the factorisation face.

1a *(medium risk; in the minimum viable thesis)* Relate the equational theory of $\equiv_2$ to the normalisation-box equational theories for causal Bayesian networks [75, 5]. *Done:* a coincidence theorem on the programs definable from causal Bayesian networks, or a separating example. *Why it may succeed:* both are equational theories over the same string-diagram semantics, differing in whether equations are indexed by programs or by graphs.

1b *(low risk)* Settle whether $\equiv_3$ is strictly finer than $\equiv_3^{2\text{-world}}$. *Done:* a separating pair of programs or a collapse proof.

1c *(medium risk)* The factorisation reading: show that the conditional independences of a program's Markov-category denotation, via Fritz and Klingler's synthetic d-separation [53], are those implied by $\equiv_2$-invariance under interventions, and characterise the gap (interventional independence is strictly stronger than conditional independence in general). *Done:* the implication proved and the gap characterised by examples. *Why it matters:* it is the precise sense in which the intervention face and the factorisation face are two readings of one semantics.

Work package 2: cycles and feedback.

2a *(medium risk; in the minimum viable thesis)* Finite cyclic models with unique solutions: add a fixpoint construct to L_2 whose denotation is Bongers et al.'s solution function on the class of finite models with unique solutions; define interventions on fixpoint-bound variables by cutting and re-solving; extend the grade and prove adequacy. *Done:* the extension, its semantics, adequacy, and Worked example 2 reproduced. *Why it may succeed:* the solution-function semantics is already a fixpoint; the question is which construct denotes it.

2b *(high risk)* The general case: existence and uniqueness of solutions as a semantic side condition; full abstraction for the cyclic language; the relation to recursion and laziness in the host language, since lazy evaluation is how OmegaC implements its retroactive interventions. *Done:* full abstraction on the restricted class, or a precise obstruction.

Work package 3: relations between models.

3a *(medium risk)* Causal abstraction as a program transformation: a coarsening map between L_2 programs that commutes with interventions, compared with the categorical abstraction line [118, 90, 50]. *Done:* a definition at the program level, a proof that it specialises to the model-level definitions on compiled SCMs, and a worked example (two variables coarsened into one).

3b *(medium risk)* The Ibeling–Icard correspondence: show that each of their three logics induces, by indistinguishability under all sentences, exactly the corresponding contextual equivalence on compiled SCMs. *Done:* the correspondence theorem or a separation. *Why it may succeed:* both hierarchies are built from Pearl's operations; they organise queries and models respectively, and indistinguishability is the standard way to turn one into the other.

Work package 4: a practical language.

4a *(low risk; in the minimum viable thesis)* Semantics for existing languages in L_{HO}: a denotational model of OmegaC's core, in which its lazy-closure interventions are shown to compute do, and an account of ChiRho's multi-world values as L_{HO} twinnings. *Done:* the translations and their correctness statements, with the paper's examples reproduced.

4b *(medium risk)* A vectorised implementation over a Python host with efficient inference, keeping as much of L_{HO}'s typing as the host allows. *Done:* the implementation, the paper's examples at realistic scale, and a comparison with ChiRho on a small benchmark set. [FILL: benchmark set to be agreed with supervisor]

Work package 5 (outlook): the signalling face. The quantum communication work of §4.3, whose completeness theorem I expect to close and submit this year, is the second place where causal structure appears as denotational data, as the order of I/O events. The signalling reading that Part

I refers to is the following statement, which I expect to follow by structural induction but do not pursue in this plan:

Conjecture 1. *For every closed term t of $\mathsf{IO} \otimes \mathsf{Q}$ and every trace σ, the component $[\![t]\!]_\sigma$ is a probabilistic comb with slot order σ, and the family $([\![t]\!]_\sigma)_\sigma$ is a variable-length tester: for each σ, the sum of the components over all extensions of σ, with the later wires discarded, is a deterministic comb with slot order σ.*

The hypothesis that an intervention label is a send/receive pair with the intervener as environment, so that $\equiv_2$ is operational equivalence under a class of streams, would make the interventional semantics a fragment of the communication one; I state it as the thesis's outlook and will check it on paper, but it is not a work package of this plan.

10 Evaluation

The thesis is theoretical with an executable component. Theorems are evaluated by the "done" criteria above: coincidence or separation of equational theories, adequacy and full abstraction for the cyclic language, specialisation theorems for abstraction and for the logical correspondence, and correctness of the translations of OmegaC and ChiRho. The implementation is evaluated by reproducing the textbook causal-inference examples (mediation, instrumental variables, feedback) at realistic scale and by a comparison with ChiRho on an agreed benchmark set. The Haskell implementation [139] is extended with every construct so that every definition remains executable.

11 Impact and risks

If successful, the thesis gives denotational foundations for causal probabilistic programming, and hence for causal machine-learning languages, in which interventions are mathematical operations rather than evaluation strategies, with the first program-level treatment of cyclic models and of abstraction, and a practical language whose semantics is the specification of its implementation.

Item	Risk	Fallback
1a equational theories	medium	a separating example is itself a result
1b strictness	low	—
1c factorisation reading	medium	the implication alone, with the gap by example
2a cycles, restricted class	medium	finite models with unique solutions only
2b cycles, general	high	a precise obstruction
3a abstraction	medium	the compiled-SCM case only
3b logical correspondence	medium	one direction
4a semantics for existing languages	low	OmegaC alone
4b implementation	medium	the examples without the benchmark comparison

Table 2: Risks and fallbacks.

The generic risk is that the extensions are closer to existing lines than the base result was. The mitigation is the *minimum viable thesis*, WP 1a, WP 2a and WP 4a, every item rated low or medium and each a program-level treatment of something that exists only at the model level; the originality lies in the program-level account, not in the choice of topics.

12 Plan of work, milestones, and training

I am in my fourth term. The baseline schedule is the minimum viable thesis; conditional items are marked.

Dissemination targets are POPL, PLDI, LICS, MFPS and UAI. The thesis will be organised as: 1 Introduction; 2 Background (Part I of this document); 3 Causal probabilistic programs:

Term	Milestones
MT26 (4)	Transfer of status. POPL revision. Completeness manuscript for $\mathsf{IO} \otimes \mathsf{Q}$ (work to date) closed and submitted. Reading milestone M1: normalisation boxes and the equational principles for causal Bayesian networks; Bongers et al. on cyclic models in depth.
HT27 (5)	WP 1a, WP 1b. WP 4a: OmegaC core. *Deliverable:* note on the equational theories.
TT27 (6)	WP 2a. WP 1c. *Deliverable:* paper on cycles and the equational theory.
MT27–HT28 (7–8)	WP 3a, WP 3b. WP 4b begun. **Confirmation of Status in HT28**, with the minimum viable thesis in hand.
TT28 (9)	Buffer for Confirmation of Status. WP 2b. *Deliverable:* paper on abstraction and the logical correspondence.
MT28 (10)	WP 4b completed and evaluated. The outlook hypothesis of WP 5 checked on paper.
HT29–TT29 (11–12)	Consolidation. Thesis writing; submission by the end of TT29.

the hierarchy, contextual equivalence and the graded semantics [FILL: delineate your contribution from coauthors']; 4 The equational theory and the factorisation face; 5 Cycles; 6 Abstraction and logics; 7 A practical language; 8 Outlook: order and signalling in quantum communication, with the completeness theorem as work to date [FILL: confirm MSc reuse rules with supervisor]; 9 Conclusion.

Training and engagement. [FILL: required by the assessors' criteria: seminars attended (PL group, quantum group); courses audited; the PLanQC 2025 talk, with a paragraph on it; teaching; planned activities for the coming year].

13 Statement of the subject of the thesis

The semantics of causal probabilistic programming: contextual equivalence as the definition of causal structure, extended to higher order, cycles and abstraction, and made executable.

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